1
changing bounds
0<=y<=1
5y<=x<=5
0<=x<=5
0<=y<=x/5
integrate(0-5)(integrate(0-x/5)(2e^x^2)dydx
integrate(0-5)(2e^x^2y)from(0-x/5)dx
integrate(0-5)(2e^x^2x/5)dx
e^x^2/5from(0-5)
e^25/5

2
integral(0-2)(integrat(0-1.5x)(6-3x-2y)dydx
integral(0-2)6y-3xy-y^2from(0-1.5x)dx
integral(0-2)9x-4.5x^2+2.25x^2dx
9/2x^2-4.5/3x^3+2.25/3x^3from(0-2)
18/2-8*4.5/3+8*2.25/3

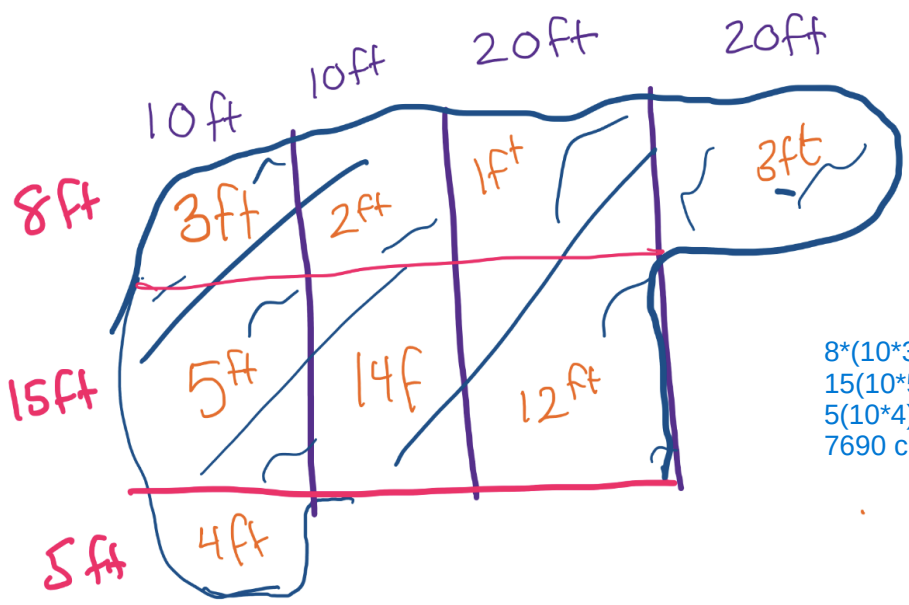
Worksheet 5, 15.1-2

Complete in Discussion

1. Evaluate
$$\int_0^1 \int_{5y}^5 2e^{x^2}\,dx\,dy$$

2. Find the volume of the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$ and the plane $3x + 2y + z = 6$.

3. The residents of Frog lake would like to determine the volume of water in their lake. They separate it into several rectangles. Then in each rectangle, they take a measurement of the depth. Use the length and depth information below to find the volume of the lake.



8*(10*3+10*2+20*1+20*3)+
15(10*5+10*14+20*12)+
5(10*4)=
7690 cubic feet

Extra Quiz/Exam Practice

4. Evaluate the double integral.
$$\iint_D x\sqrt{y^2 - x^2}\,dA, D = \{(x, y)|0 \le y \le 1, 0 \le x \le y\}$$

5. Evaluate the iterated integral below

$$\int_0^2 \int_2^3 \frac{xe^x}{y}\, dy\, dx.$$

6. Evaluate the integral:

$$\int_0^3 \int_{x^2}^9 x\cos(y^2)\, dy\, dx$$

7. Evaluate

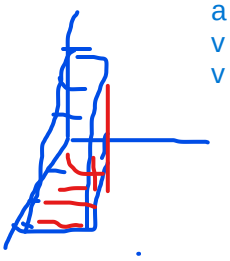
$$\int_a^b \int_c^d \frac{2xy^2}{x^2+1}\, dy\, dx.$$

By splitting into the product of two single integrals. When can you split an integral like this?

Challenge Questions

8. Evaluate the integral below in each of the following ways.

a
v = .5*l*w*h
v = .5*3*3*1 = 9/2



$$\iint_R (3-y)dA \qquad R = [0,1]\times[0,3]$$

b
integral(0-3)(integral(0-1)(3-y)dxdy
integral(0-3)(3x-yx)from(0-1)dx
integral(0-3)(3-y)dy
(3y-y^2/2)from(0-3)
9-9/2 = 9/2

(a) Geometrically. (Sketch a graph and determine the volume)

(b) Using calculus. (Evaluate the iterated integral.)

9. We want to evaluate

$$\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3}\, dx\, dy.$$

A sketch of the solution is below. Identify one issue with this evaluation.(Hint: see section 7.8 in Stewart). Then carry out a similar method to find

$$\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3}\, dy\, dx.$$

(You do not need to fix the 7.8 issue you identified.) You should get $\frac{1}{2}$ instead of $-\frac{1}{2}$ Why does this not follow Fubini's Theorem? (Why doesn't it meet the hypotheses?)

It does not follow the theorem because it is
not continuous over the domain of the
iterated integral

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integral(0-1)(integral(0-1)((x-y)/(x+y)^3)dydx
u = x+y, y = u-x du=dy
integral(0-1)(integral(0-1)((x-u+x)/(u^3)dydx
integral(0-1)(integral(0-1)((2x-u)/(u^3)dydx
integral(0-1)(-x/u^2+1/u)from(0-1)dx
integral(0-1)(1/(x+1)^2)dx
-1/(x+1)from(0-1)
-.5+1=.5

$$
\begin{aligned}
\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3}\,dx\,dy &= \int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3}\,dx \right) dy \\
&= \int_0^1 \left(\int_{x=0}^{x=1} \frac{(u-y)-y}{(u)^3}\,du \right) dy \text{ where } u = x+y, x = u-y, dx = du \\
&= \int_0^1 \left(\int_{x=0}^{x=1} \frac{u-2y}{u^3}\,du \right) dy \qquad \text{the mistake is when solving for u} \\
&= \int_0^1 \left(\int_{x=0}^{x=1} \frac{1}{u^2} + \frac{-2y}{u^3}\,du \right) dy \\
&= \int_0^1 \left(\frac{-1}{u} + \frac{y}{u^2}\Bigg|_{x=0}^{x=1} \right) dy \\
&= \int_0^1 \left(\frac{-1}{x+y} + \frac{y}{(x+y)^2}\Bigg|_{x=0}^{x=1} \right) dy \\
&= \int_0^1 \left(\frac{-1}{1+y} + \frac{y}{(1+y)^2} - \left(\frac{-1}{y} + \frac{y}{(y)^2} \right) \right) dy \\
&= \int_0^1 \left(\frac{-1(1+y)}{(1+y)^2} + \frac{y}{(1+y)^2} \right) dy \\
&= \int_0^1 \frac{-1}{(1+y)^2}\,dy \\
&= \frac{1}{(1+y)}\Bigg|_0^1 \\
&= \frac{1}{(1+1)} - \frac{1}{1} \\
&= \frac{-1}{2}.
\end{aligned}
$$

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